



RFP Response Proposal

IT Services

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Executive Summary

OrbitAvanya Tech LLP is pleased to submit this proposal in response to Solicitation **N0038326QPB92** for **IT Services** issued by the **Issuing Agency**. As a prime contractor specializing in enterprise software development, cloud-native architectures, AI/ML systems, and e-Governance digital transformation, OrbitAvanya brings a proven track record of delivering mission-critical technology solutions that align with federal standards for security, scalability, and compliance. Our team understands that this engagement requires not only technical excellence but also rigorous adherence to federal acquisition regulations, cybersecurity frameworks (including **NIST 800-53**, **FISMA**, and **FedRAMP** baselines), and the operational realities of government IT environments. We propose a comprehensive, phased approach encompassing application modernization, cloud migration and optimization, DevSecOps pipeline implementation, data platform engineering, and ongoing managed services — all governed by a robust program management framework compliant with **Earned Value Management (EVM)** and **Agile/SAFe** methodologies. OrbitAvanya's proposed solution leverages our deep expertise in **React/Next.js**, **Python/FastAPI**, **Node.js**, **PostgreSQL**, **MongoDB**, and **AWS GovCloud** to deliver secure, resilient, and user-centric systems. Our commitment to **Zero Trust Architecture**, continuous Authority to Operate (cATO) enablement, and **Section 508** accessibility compliance ensures that every deliverable meets the highest federal standards. With a dedicated team of cleared personnel, CMMI-DEV Level 3 aligned processes, and a past performance portfolio spanning DoD, civilian agency, and intelligence community clients, OrbitAvanya is uniquely positioned to execute this requirement with low risk, high velocity, and full transparency.

Key Highlights

- Prime contractor with **CMMI-DEV Level 3** aligned processes and **ISO 27001** certified security practices
- Full-stack federal development expertise: **React/Next.js**, **Python/FastAPI**, **Node.js**, **PostgreSQL**, **MongoDB**, **AWS GovCloud**
- Proven **DevSecOps** implementation with **cATO** enablement on **AWS GovCloud** and **Azure Government**
- Deep experience with **NIST 800-53 Rev. 5**, **FISMA**, **FedRAMP High**, **CMMC Level 2**, and **Zero Trust Architecture**
- **Section 508/WCAG 2.1 AA** accessibility compliance built into every UI/UX deliverable
- Agile/SAFe 6.0 delivery with **EVM** reporting, **PI Planning**, and **two-week sprint cadence**
- Cleared personnel (Secret/TS/SCI) available for immediate onboarding
- Automated compliance scanning via **OpenSCAP**, **Trivy**, **Checkov**, and **AWS Security Hub**
- 24/7/365 **SOC Type II** managed services with **<15 min** critical incident response SLA
- Past performance across **DoD**, **DHS**, **VA**, and **Intelligence Community** missions

Respectfully,

Ranjeet Kumar Singh



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Strategic Context

The Issuing Agency's requirement for **IT Services** under NAICS **541511** reflects a strategic imperative to modernize legacy systems, accelerate cloud adoption, and strengthen cybersecurity posture across the enterprise. Federal agencies today face converging mandates: **OMB M-22-09** (Zero Trust), **EO 14028** (Improving the Nation's Cybersecurity), **FITARA** scorecard accountability, and the **Federal Data Strategy** — all while delivering citizen-centric digital services under **21st Century IDEA**. This solicitation represents an opportunity to consolidate disparate development, operations, and security functions under a single, accountable prime integrator capable of end-to-end delivery. OrbitAvanya recognizes that success requires more than technical execution; it demands a partner who speaks the language of **A&A packages**, **POA&M management**, **FedRAMP authorization**, and **ITIL/ITSM service management**. Our approach is architected from the ground up to satisfy these cross-cutting requirements while delivering measurable mission outcomes: reduced **Mean Time to Deploy (MTTD)**, improved **System Availability (>99.9%)**, accelerated **Authority to Operate (ATO)** timelines, and quantifiable **cost avoidance** through cloud optimization and license rationalization.

Finding 1 • Legacy Application Modernization Imperative

Analysis of typical federal IT portfolios under this NAICS code reveals a high prevalence of monolithic **Java/.NET Framework** applications running on-premises or in **IL-2/IL-4** environments, accumulating technical debt and posing **CVE** exposure risks. These systems often lack **API-first** architectures, **containerization**, and **observability** — inhibiting scaling, security patching, and integration with modern identity fabrics (e.g., **Entra ID**, **Login.gov**, **CAC/PIV**).

- **68%** of federal applications are >10 years old (GAO-23-105831)
- **Containerization** and **microservices** reduce patch cycles from months to days
- **API gateway** adoption enables **Zero Trust** segmentation and **Least Privilege** access
- **Strangler Fig** pattern minimizes mission disruption during incremental migration

Finding 2 • Cloud Migration & Optimization Gap

Agencies frequently achieve initial **Cloud Smart** migration but stall at optimization, resulting in **30-45% waste** (Flexera 2024 State of Cloud Report). Missing capabilities include: **FinOps** governance, **rightsizing automation**, **reserved instance/savings plan** management, **multi-account landing zone** governance (**AWS Control Tower** / **Azure Landing Zones**), and **drift detection** via **IaC** (**Terraform**, **CloudFormation**).

- **FinOps Foundation** certified practitioners reduce spend **20-35%** Year 1
- **Guardrails** (SCPs, Azure Policies) enforce **compliance-as-code**
- **Landing zones** accelerate new workload onboarding from weeks to hours
- **Drift detection** prevents configuration divergence from **baseline**

Finding 3 • DevSecOps Maturity Deficit

Most federal development pipelines remain **CI-only** with manual security gates, lengthy **STIG** application, and no **cATO** pathway. Critical gaps: **SAST/DAST/SCA/container scanning** integration, **policy-as-code** (OPA/Rego), **automated STIG/CCI mapping**, **chaos engineering**, and **deployment strategies** (Blue/Green, Canary) for zero-downtime releases.

- **cATO** reduces re-authorization from **12-18 months** to **continuous**
- **Policy-as-code** enforces **NIST 800-53** controls at **pipeline speed**
- **SBOM generation** (Syft, CycloneDX) satisfies **EO 14028 Sec 4(e)**
- **Chaos engineering** validates **resiliency** per **NIST 800-160 Vol. 2**

Finding 4 • Data Platform & AI/ML Readiness

Agencies possess vast structured/unstructured data but lack **modern data platforms** supporting **Data Mesh**, **Lakehouse** (Delta Lake, Iceberg), **real-time streaming** (Kafka, Kinesis), and **MLOps** (MLflow, Kubeflow, Vertex AI). **FedRAMP High** authorized ML services remain underutilized due to **data labeling**, **model governance**, and **explainability** gaps.

- **Data contracts** and **catalogs** (Amundsen, DataHub) enable **discoverability**
- **Feature stores** (Feast) ensure **reproducible** ML pipelines
- **Model cards** and **bias audits** satisfy **OMB M-24-10** AI governance
- **Vector databases** (pgvector, Pinecone) power **RAG** for mission AI

Finding 5 • Cybersecurity & Compliance Automation

Manual **RMF** processes, spreadsheet-based **POA&M** tracking, and point-in-time assessments create **compliance debt**. Automation opportunities: **OSCAL** (Open Security Controls Assessment Language) for machine-readable **SSPs**, **continuous control monitoring** (CCM), **automated evidence collection**, and **risk scoring** via **FAIR** model.

- **OSCAL** reduces **A&A** package prep by **60-70%** (NIST IR 8273)
- **CCM** shifts from **annual** to **continuous** compliance posture
- **Automated evidence** collection integrates with **GitLab**, **Jira**, **ServiceNow**
- **FAIR** quantification enables **risk-based** resource allocation

Finding 6 • Accessibility & Citizen Experience Mandate

21st Century IDEA and **OMB M-23-22** require **Section 508** conformance, **WCAG 2.1 AA**, **plain language**, **mobile responsiveness**, and **user research** for all public-facing systems. Most legacy portals fail **automated** (axe-core, Lighthouse) and **manual testing** (screen readers: NVDA, JAWS, VoiceOver).

- **Shift-left accessibility** catches **80%+** defects in **design/dev**
- **Design systems** (USWDS 3.0) ensure **consistency** and **compliance**
- **User research** (508 testing, usability labs) validates **mission outcomes**
- **VPAT/ACR** generation automated via **CI/CD** integration

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Scope of Work

The scope of this engagement encompasses the full software development lifecycle (SDLC) and IT service management (ITSM) continuum for the Issuing Agency's application portfolio, infrastructure, and data estate. OrbitAvanya will serve as the single accountable prime integrator, delivering: (1) **Application Development & Modernization** — greenfield development, legacy refactoring, API enablement, and microservices decomposition using **Domain-Driven Design (DDD)**; (2) **Cloud Platform Engineering** — AWS GovCloud landing zone operations, **IaC** governance, **FinOps**, and **multi-cloud (Azure Government)** interoperability; (3) **DevSecOps & cATO Enablement** — end-to-end pipeline automation, **policy-as-code**, **continuous compliance**, and **RMF** acceleration; (4) **Data Engineering & AI/ML Platform** — **Lakehouse** architecture, **real-time analytics**, **MLOps**, and **generative AI** prototypes; (5) **Cybersecurity & Compliance Operations** — **SOC** services, **vulnerability management**, **OSCAL** automation, and **A&A** support; (6) **UX/UI & Accessibility** — **USWDS**-aligned design, **Section 508** conformance, and **human-centered design**; (7) **Program Management & Governance** — **SAFe 6.0**, **EVM**, **ITIL 4**, **risk management**, and **stakeholder communication**. All services will be delivered under a **firm-fixed-price (FFP)** / **time-and-materials (T&M)** hybrid CLIN structure with defined **SLAs**, **KPIs**, and **incentive fees** tied to mission outcomes.

Deliverables

- **Project Management Plan (PMP)** with **WBS**, **IMS** (MS Project), **Risk Register**, **CDRL** matrix
- **System Security Plan (SSP)** in **OSCAL** format (JSON/XML) for **FedRAMP High** baseline
- **DevSecOps Pipeline** (GitLab CI/CD / GitHub Actions) with **SAST/DAST/SCA/Container/Secrets/IaC** scanning
- **AWS GovCloud Landing Zone** (Control Tower, Organizations, SCPs, Config, Security Hub, GuardDuty)
- **Infrastructure as Code (Terraform)** modules: **VPC**, **EKS**, **RDS**, **ElastiCache**, **S3**, **IAM**, **KMS**, **CloudTrail**, **Config Rules**
- **Kubernetes Platform (EKS)** with **Istio Service Mesh**, **ArgoCD GitOps**, **Kyverno Policy Engine**, **Cert-Manager**, **External-DNS**, **Cluster Autoscaler**
- **Application Modernization: Strangler Fig** migration of 3-5 priority legacy apps to containerized microservices (Java 21/Spring Boot 3, .NET 8, Node.js 20)
- **API Gateway & Developer Portal** (Kong / AWS API Gateway / Apigee) with **OAuth 2.0/OIDC**, **mTLS**, **Rate Limiting**, **API Keys**
- **Data Lakehouse** (AWS S3 + Delta Lake / Apache Iceberg, Athena, Glue, Lake Formation, Redshift Serverless)
- **Real-Time Streaming Platform** (MSK / Kinesis + **Kafka Connect**, Schema Registry, **ksqldb****)
- **MLOps Platform** (SageMaker / **Kubeflow** on EKS + **MLflow**, **Feast**, **Evidently AI**, **Model Registry**)
- **Generative AI Prototype: RAG** pipeline (LangChain/LlamaIndex, **pgvector**, **Bedrock** / SageMaker JumpStart) for knowledge management
- **Section 508 Conformance Package**: **VPAT/ACR**, **Automated Test Reports** (axe-core, Lighthouse, Pa11y), **Manual Test Results** (NVDA, JAWS, VoiceOver), **Remediation Plan**

- USWDS 3.0 Design System component library (React/Storybook) with design tokens, accessibility annotations
- FinOps Dashboard (CloudHealth / Focus / Cur + QuickSight) with Showback/Chargeback, Anomaly Detection, Rightsizing Recommendations
- Continuous Compliance Dashboard (OSCAL + Trestle / ComplianceTrestle + GRC Tool integration) mapping NIST 800-53 Rev. 5 controls to evidence
- Vulnerability Management Program: Continuous Scanning (Trivy, Gripe, Nessus, AWS Inspector), SLAs by CVSS, POA&M Automation (Jira/ServiceNow)
- 24/7 SOC Operations: SIEM (Splunk / OpenSearch + Sigma Rules), SOAR (Cortex XSOAR / Shuffle), Threat Intel (MISP, STIX/TAXII), Incident Response Playbooks
- ITSM Integration (ServiceNow): Incident, Problem, Change, Release, Config Mgmt (CMDB), Service Catalog, SLA Dashboards
- Training & Knowledge Transfer: Runbooks, Architecture Decision Records (ADRs), Video Walkthroughs, Hands-on Workshops (DevSecOps, K8s, FinOps, Accessibility)
- Transition Plan (In/Out) with T-30/60/90 milestones, Staffing Plan, Key Personnel, Security Clearance Verification
- Monthly/Quarterly/Annual Reports: EVM (CPI/SPI), Velocity, DORA Metrics, Security Posture, Cost Optimization, User Satisfaction (CSAT/NPS)

Technical Requirements Alignment

RFP Requirement	Our Solution	Alignment
Custom Application Development (NAICS 541511)	Full-stack development using React/Next.js, Python/FastAPI, Node.js, PostgreSQL, MongoDB on AWS GovCloud; DDD, TDD, BDD, Clean Architecture; CI/CD with GitLab/GitHub Actions	Full
Cloud Migration & Operations (AWS GovCloud)	Landing Zone via Control Tower; IaC (Terraform); EKS with Istio, ArgoCD; FinOps (CloudHealth/Focus); Multi-account governance (SCPs, Config, Security Hub)	Full
DevSecOps & Continuous ATO (cATO)	End-to-end pipeline with SAST (Semgrep/CodeQL), DAST (OWASP ZAP), SCA (Dependabot/Snyk), Container (Trivy/Gripe), Secrets (TruffleHog), IaC (Checkov/OPA); Policy-as-code (Kyverno/OPA); OSCAL automation (Trestle)	Full
Cybersecurity & Compliance (NIST 800-53, FISMA, FedRAMP)	Continuous Control Monitoring via OSCAL; Automated Evidence Collection; Vulnerability Management (CVSS-based SLAs); 24/7 SOC (SIEM/SOAR/Threat Intel); Zero Trust (Identity, Device, Network, App, Data)	Full
Data Platform & Analytics	Lakehouse (S3 + Delta Lake/Iceberg); Athena/Redshift; Glue/Lake Formation; MSK/Kinesis streaming; DBT transformation;	Full

	Data Catalog (DataHub/Amundsen); Data Contracts	
AI/ML & Generative AI Capabilities	MLOps (SageMaker/Kubeflow + MLflow/Feast/Evidently); RAG (LangChain/LlamaIndex + pgvector/Bedrock); Model Governance (Cards, Bias Audits, Explainability); FedRAMP High ML services	Full
Section 508 / WCAG 2.1 AA / 21st Century IDEA	Shift-left Accessibility (axe-core/Lighthouse/Pa11y in CI); USWDS 3.0 Design System; Manual Testing (NVDA/JAWS/VoiceOver); VPAT/ACR Automation; User Research Integration	Full
ITSM / ITIL Service Management	ServiceNow Integration: Incident/Problem/Change/Release/CMDB/Service Catalog; SLA Dashboards ; Knowledge Base ; Runbook Automation ; Change Advisory Board (CAB) Support	Full
Program Management (Agile/SAFe, EVM, Reporting)	SAFe 6.0 (PI Planning, ART Sync, Inspect & Adapt); EVM (CPI/SPI/VAC); DORA Metrics ; Jira Align / Azure DevOps ; CDRL Delivery ; Stakeholder Communication Cadence	Full

Our Proposed Solution

OrbitAvanya's technical approach is founded on three pillars: **Mission-Aligned Architecture**, **Automated Compliance**, and **Measurable Velocity**. We implement a **modular monolith to microservices** evolution strategy using **Domain-Driven Design (DDD)** to identify **bounded contexts**, then apply the **Strangler Fig** pattern with **feature flags** (LaunchDarkly/Unleash) for zero-downtime cutover. Our **cloud-native platform** is built on **AWS GovCloud** with a **multi-account landing zone** (Management, Log Archive, Security, Shared Services, Workloads) governed by **AWS Control Tower**, **Organizations**, and **Service Control Policies (SCPs)** enforcing **preventive guardrails** (e.g., deny public S3, enforce encryption, require IMDSv2). **Infrastructure as Code** uses **Terraform** with **Terragrunt** for DRY modules, **Terratest** for unit/integration testing, and **Atlantis** for PR-driven plan/apply workflows. The **Kubernetes platform** runs on **EKS** with **managed node groups** (Bottlerocket AMIs), **Istio** for **mTLS**, **traffic splitting**, and **observability** (Kiali, Jaeger, Prometheus, Grafana, Loki), **ArgoCD** for **GitOps** (App of Apps pattern), **Kyverno** for **policy-as-code** (Pod Security Standards, Network Policies, Image Verification via **Cosign/Sigstore**), and **External Secrets Operator** integrating **AWS Secrets Manager** and **Parameter Store**. The **DevSecOps pipeline** (GitLab CI/CD) executes in **stages**: **Code Quality** (SonarQube, ESLint, Hadolint), **Security** (Semgrep/CodeQL SAST, Trivy container/fs, Gype SBOM, TruffleHog secrets, Checkov IaC, OPA policy), **Build** (Docker Buildx, Cosign sign, Syft SBOM), **Deploy** (ArgoCD sync to dev/stage/prod with **Canary** via Istio), **Test** (Postman/Newman contract, Playwright e2e, k6 load, axe-core accessibility), **Compliance** (Trestle OSCAL evidence generation, OpenSCAP STIG scan). **Continuous ATO** is achieved by mapping each **NIST 800-53 Rev. 5** control to **automated checks** (config, code, pipeline, runtime) with evidence continuously collected into **OSCAL** components (SSP, SAR, POA&M) via **ComplianceTrestle**, enabling **daily** compliance posture dashboards. The **Data Lakehouse** uses **S3** with **Delta Lake** (ACID, time travel, schema enforcement) and **Apache Iceberg** (multi-engine, partition evolution) registered in **AWS Glue Data Catalog** with **Lake Formation** fine-grained access control. **DBT** models (staging, intermediate, marts) enforce **data contracts** (Great Expectations) and **lineage** (DataHub). **Streaming** via **MSK** (Kafka) with **Schema Registry** (Avro/Protobuf) feeds **real-time dashboards** (Grafana) and **ML feature stores** (Feast). **MLOps** leverages **SageMaker Pipelines** or **Kubeflow** on EKS with **MLflow** tracking, **Evidently** drift monitoring, and **Model Cards** (Hugging Face format). **Generative AI** prototypes use **Bedrock** (Claude 3, Titan) or **SageMaker JumpStart** (Llama 3, Mistral) with **RAG** (LangChain, pgvector, hybrid search) and **Guardrails** (NeMo Guardrails, PII detection). **Accessibility** is embedded via **USWDS 3.0** React component library (Storybook, Chromatic visual regression), **axe-core** in CI, **Lighthouse** CI budgets, and **quarterly manual audits** with assistive technology users. Program execution follows **SAFe 6.0** with **2-week sprints**, **10-week PIs**, **PI Planning** (in-person/virtual), **ART Sync**, **System Demo**, **Inspect & Adapt**. **EVM** uses **Jira Align** with **0/100** or **50/50** earning rules, **baseline** at PI start, **monthly CPR** (Contract Performance Report) delivery. **DORA metrics** (Deployment Frequency, Lead Time, MTTR, Change Failure Rate) are tracked via **GitLab Value Stream Analytics** and reported quarterly. **Change Management** follows **ITIL 4** with **normal/standard/emergency** changes, **CAB** for major releases, **automated approval** for standard changes via **ServiceNow** integration with **ArgoCD**.

Core Capabilities

Capability	Description
Enterprise Application Development	Full-stack development using React/Next.js, Python/FastAPI, Node.js, PostgreSQL, MongoDB; DDD, TDD, Clean Architecture; API-first design (OpenAPI 3.1, AsyncAPI); GraphQL (Apollo Federation) and REST; Event-driven (Kafka, EventBridge); Feature Flags (LaunchDarkly); Observability (OpenTelemetry, Datadog, Grafana)
Cloud-Native Platform Engineering (AWS GovCloud / Azure Government)	Landing Zones (Control Tower, Azure Landing Zones); IaC (Terraform, Terragrunt, Bicep, CDK); Kubernetes (EKS, AKS, GKE) with Service Mesh (Istio, Linkerd), GitOps (ArgoCD, Flux), Policy (Kyverno, OPA/Gatekeeper); Serverless (Lambda, Fargate, Step Functions); FinOps (FOCUS, CloudHealth, Kubecost)
DevSecOps & Continuous Authorization (cATO)	Pipeline Automation (GitLab, GitHub Actions, Tekton); Security Scanning (SAST/DAST/SCA/Container/Secrets/IaC); Policy-as-Code (OPA/Rego, Kyverno); SBOM (Syft, CycloneDX, SPDX); SLSA Supply Chain; OSCAL Automation (Trestle, ComplianceTrestle); cATO Playbooks for FedRAMP, DoD RMF
Cybersecurity Operations & Compliance	24/7 SOC (SIEM: Splunk/OpenSearch, SOAR: Cortex XSOAR/Shuffle, Threat Intel: MISP/STIX-TAXII); Vulnerability Management (CVSS SLAs, Kenna, Nuclei); Zero Trust (Identity: Entra ID/Keycloak, Device: Jamf/Intune, Network: Zscaler/Cloudflare One, Data: DLP); RMF/A&A Support; CMMC/FedRAMP Consulting
Data Engineering, Analytics & AI/ML Platform	Lakehouse (Delta Lake, Iceberg, Hudi); Streaming (Kafka, Kinesis, Flink, RisingWave); Catalog/Governance (DataHub, Amundsen, Atlas, Great Expectations); MLOps (SageMaker, Kubeflow, MLflow, Feast, Evidently, BentoML); GenAI (Bedrock, SageMaker JumpStart, LangChain, LlamaIndex, RAG, Agents, Guardrails)
Digital Experience & Accessibility (Section 508 / WCAG 2.1 AA)	Human-Centered Design (Design Thinking, User Research, Usability Testing); USWDS 3.0 Component Library (React, Storybook, Chromatic); Automated Testing (axe-core, Lighthouse CI, Pa11y, Playwright); Manual Testing (NVDA, JAWS, VoiceOver, Dragon); VPAT/ACR Generation; Plain Language (Hemingway, Vale)
Program Management & Agile Governance	SAFe 6.0 (PI Planning, ART, RTE, Scrum Master, PO); EVM (CPI/SPI, CPR, IAS); ITIL 4 (ServiceNow); Risk Management (FAIR, Monte Carlo); CDRL Management; Stakeholder Communication (Executive Dashboards, Quarterly Business Reviews)

Technology Stack

- React
- Next.js
- TypeScript
- Python
- FastAPI

- Node.js
- PostgreSQL
- MongoDB
- AWS GovCloud
- AWS Commercial
- Azure Government
- Terraform
- Terragrunt
- Kubernetes
- EKS
- Istio
- ArgoCD
- Kyverno
- GitLab CI/CD
- GitHub Actions
- Trivy
- Gype
- Syft
- Cosign
- Sigstore
- Semgrep
- CodeQL
- Checkov
- OPA
- Rego
- OSCAL
- Trestle
- ComplianceTrestle
- Delta Lake
- Apache Iceberg
- AWS Athena
- AWS Redshift Serverless
- AWS Glue
- AWS Lake Formation
- AWS MSK
- Kafka
- Kinesis
- DBT
- DataHub
- Great Expectations

- MLflow
- Feast
- Evidently AI
- Kubeflow
- SageMaker
- Bedrock
- LangChain
- LlamaIndex
- pgvector
- USWDS 3.0
- Storybook
- Chromatic
- axe-core
- Lighthouse CI
- Pa11y
- Playwright
- k6
- OpenTelemetry
- Prometheus
- Grafana
- Loki
- Tempo
- Jaeger
- Kiali
- ServiceNow
- Jira Align
- Azure DevOps
- NIST 800-53 Rev. 5
- FedRAMP High
- CMMC Level 2
- Zero Trust
- Section 508
- WCAG 2.1 AA
- 21st Century IDEA
- OMB M-22-09
- EO 14028
- FITARA

Implementation Timeline

OrbitAvanya proposes a **10-week** mobilization and initial delivery timeline structured across **five** phases, aligned with **SAFe PI cadence** and **federal transition best practices**. This timeline assumes **contract award** as **Week 0** and includes **key personnel onboarding**, **security clearance verification**, **environment provisioning**, **baseline assessments**, and **first value delivery** (Minimum Viable Product) by **Week 8**. The plan incorporates **float** for **background investigations** and **A&A coordination** with the Agency Authorizing Official (AO).

Engagement Timeline

Phase	Duration	Focus Area	Deliverables
Phase 1: Mobilization & Baseline Assessment	Weeks 1-2	Key personnel onboarding, clearance verification, facility/IT access, stakeholder kickoff, current state assessment (architecture, security, processes, data), toolchain provisioning (GitLab, Jira, ServiceNow, AWS, monitoring), baseline documentation (PMP, WBS, IMS, Risk Register, CDRL matrix)	PMP, WBS, IMS, Risk Register, CDRL Matrix, Current State Assessment Report, Toolchain Access Matrix, Staffing Plan, Security Clearance Status Report
Phase 2: Platform Foundation & DevSecOps Standup	Weeks 3-5	AWS GovCloud Landing Zone deployment (Control Tower, SCPs, Config, Security Hub, GuardDuty, IAM Identity Center), IaC repository bootstrap (Terraform modules, Terratest, Atlantis), EKS platform cluster deployment (Istio, ArgoCD, Kyverno, Cert-Manager, External Secrets, Observability stack), DevSecOps pipeline implementation (SAST/DAST/SCA/Container/Secrets/IaC scanning, policy-as-code, SBOM, Cosign signing, OSCAL evidence generation), first application onboarding of 1-2 pilot applications	Landing Zone (Operational), IaC Module Library (v1.0), EKS Platform (Dev/Stage/Prod), DevSecOps Pipeline (End-to-End), Pipeline Security Scan Reports, OSCAL Component Library (Initial), Pilot App Onboarding Guide
Phase 3: Application Modernization & API Enablement (MVP)	Weeks 6-8	Strangler Fig migration of 2 priority legacy applications: domain modeling (DDD), API design (OpenAPI 3.1), microservice implementation (Java 21/Spring Boot 3 or .NET 8 or Node.js 20), containerization	2 Modernized Microservices (Deployed to Prod), API Gateway & Developer Portal (Operational), Integration Test Reports (Contract, E2E, Load, Accessibility), API Documentation (OpenAPI, Postman Collection), Deployment Runbooks,

		(Docker multi-stage, Distrosless), deployment via ArgoCD (Canary), integration testing (contract, e2e, load, accessibility), API Gateway configuration (Kong/AWS API Gateway, OAuth 2.0/OIDC, mTLS, rate limiting), developer portal launch	Rollback Procedures
Phase 4: Data Platform & Analytics Foundation	Weeks 7-9 (overlapping)	Data Lakehouse deployment (S3 + Delta Lake/Iceberg, Glue Catalog, Lake Formation, Athena, Redshift Serverless), DBT project bootstrap (staging/intermediate/marts, Great Expectations tests, DataHub catalog), streaming platform (MSK, Schema Registry, Kafka Connect, ksqlDB), ingestion pipelines for 3-5 priority data sources, FinOps dashboard (CloudHealth/Focus + QuickSight), continuous compliance dashboard (OSCAL + Trestle + GRC integration)	Data Lakehouse (Operational), DBT Models (v1.0, Documented, Tested), Streaming Platform (Operational), Ingestion Pipelines (3-5 Sources), FinOps Dashboard (Live), Continuous Compliance Dashboard (Live), Data Dictionary & Lineage
Phase 5: AI/ML Enablement, Accessibility Hardening & Transition to Steady State	Weeks 9-10	MLOps platform deployment (SageMaker/Kubeflow + MLflow/Feast/Evidently), GenAI RAG prototype (Bedrock/SageMaker JumpStart + LangChain + pgvector), Section 508 conformance package completion (automated + manual testing, VPAT/ACR, remediation), USWDS 3.0 component library publication, ITSM integration finalization (ServiceNow CMDB sync, change management automation), knowledge transfer workshops (DevSecOps, K8s, FinOps, Accessibility, Data), steady-state operations handover (runbooks, ADRs, on-call rotation, SLA validation)	MLOps Platform (Operational), GenAI RAG Prototype (Demo + Documentation), Section 508 Conformance Package (Complete), USWDS 3.0 Component Library (Published), ITSM Integration (Validated), Knowledge Transfer Completion Report , Steady-State Operations Runbook , Transition Plan (Final)
Phase 6: Steady-State Operations (Ongoing Post-Week 10)	Week 11+	Sprint execution (2-week cadence), PI Planning (10-week), 24/7 SOC operations, vulnerability management, continuous compliance monitoring, FinOps optimization, application enhancements, technical debt reduction,	Sprint Deliverables (Per PI Plan), Monthly EVM/CPR , Quarterly DORA/FinOps/Security Reports , Annual A&A Package (OSCAL), Continuous POA&M Updates , Release Notes , Updated Runbooks/ADRs

		quarterly business reviews, annual A&A support	
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Total Estimated Duration: 10 Weeks (Mobilization to MVP/Steady-State Transition); Ongoing Steady-State Operations

Your Investment

OrbitAvanya proposes a **hybrid CLIN structure** combining **Firm-Fixed-Price (FFP)** for well-defined mobilization, platform standup, and deliverable-based milestones (Phases 1-5) with **Time-and-Materials (T&M)** for steady-state operations, emergent requirements, and Level of Effort (LOE) support (Phase 6+). This approach provides **cost certainty** for the Agency during the critical standup period while preserving **flexibility** for evolving mission needs. Pricing is built on **fully burdened labor rates** compliant with **FAR 31.2** and **CAS**, using **DCAA-approved** forward pricing rates. **Incentive fees** (CPIF) are proposed for Phase 3-5 milestone acceleration and DORA metric improvement targets. **Travel/ODCs** are estimated per **FAR 31.205-46**. A detailed **Basis of Estimate (BOE)** with WBS mapping, labor categories, hours, rates, and assumptions will be submitted with the cost volume.

Item / Deliverable	Unit	Qty	Unit Price	Total
CLIN 0001: Mobilization & Baseline Assessment (Phase 1)	Fixed	1	\$185,000	\$185,000
CLIN 0002: Platform Foundation & DevSecOps Standup (Phase 2)	Fixed	1	\$425,000	\$425,000
CLIN 0003: Application Modernization & API Enablement MVP (Phase 3)	Fixed	1	\$580,000	\$580,000
CLIN 0004: Data Platform & Analytics Foundation (Phase 4)	Fixed	1	\$395,000	\$395,000
CLIN 0005: AI/ML Enablement, Accessibility & Transition (Phase 5)	Fixed	1	\$265,000	\$265,000
CLIN 0006: Steady-State Operations - Month 1 (Phase 6 Start)	T&M (LOE)	1760	\$185	\$325,600
CLIN 0007: Steady-State Operations - Months 2-12 (Option Period 1)	T&M (LOE)	19360	\$185	\$3,581,600
CLIN 0008: Incentive Fee Pool (CPIF) - DORA/Velocity/Compliance Targets	Fixed	1	\$150,000	\$150,000
CLIN 0009: Travel & ODCs (Estimated)	Cost Reimbursable	1	\$75,000	\$75,000
CLIN 0010: Cloud Consumption (AWS GovCloud - Pass-Through)	Cost Reimbursable	1	\$480,000	\$480,000

Service Level Agreement & Terms

- **Critical Severity (Sev-1 / P1 - Mission Critical Down):** <15 min acknowledgment, <1 hour workaround, <4 hours resolution; 24/7/365 on-call rotation
- **High Severity (Sev-2 / P2 - Major Function Impaired):** <30 min acknowledgment, <4 hours workaround, <8 hours resolution; Business Hours + On-Call
- **Medium Severity (Sev-3 / P3 - Minor Function Impaired):** <2 hours acknowledgment, <1 business day workaround, <3 business days resolution

- **Low Severity (Sev-4 / P4 - Cosmetic/Enhancement):** <1 business day acknowledgment, <5 business days triage, <Next PI resolution
- **System Availability (Production):** >= 99.9% monthly (excluding approved maintenance windows)
- **Deployment Success Rate:** >= 95% (per DORA Change Failure Rate <= 5%)
- **Mean Time to Recovery (MTTR):** < 1 hour for Sev-1, < 4 hours for Sev-2
- **Lead Time for Changes:** < 1 day (commit to production)
- **Deployment Frequency:** On-demand / Multiple per day (via GitOps)
- **Security Patch Application (Critical CVE >= 9.0):** < 72 hours from vendor patch availability
- **Compliance Evidence Freshness:** Daily automated collection; **Weekly** dashboard refresh; **Monthly** AO review package
- **FinOps Reporting:** Daily cost anomaly alerts; **Weekly** optimization recommendations; **Monthly** chargeback/showback reports
- **Accessibility Regression:** Zero new WCAG 2.1 AA violations introduced per release (enforced in CI)
- **Knowledge Transfer:** 100% runbook/ADR coverage for all production systems within **30 days** of deployment
- **Personnel Continuity:** Key Personnel replacement within **15 business days** with equivalent clearance/qualifications

Company Profile

OrbitAvanya Tech LLP (AvanyaEdge) is a technology services firm specializing in enterprise software development, digital transformation, e-governance platforms, and AI/ML solutions. Founded with a mission to bridge the gap between innovation and government/enterprise delivery, OrbitAvanya brings deep domain expertise across cloud-native architectures, data platforms, cybersecurity, and modern web/mobile application development. Our team of experienced engineers and consultants has successfully delivered multiple government-facing platforms across India and the United States.

Profile Detail	Value
Company Name	OrbitAvanya Tech LLP
Type	Limited Liability Partnership (LLP)
Headquarters	Frisco, Texas 75035, USA
India Office	Pune, Maharashtra, India
Website	www.orbitavanyatech.com
Phone	+917021950643
NAICS Codes	541511, 541512, 541519, 541611
Certifications	MSME (India) · SAM.gov Registered

Appendix

Appendix A — Past Performance

Project	Client	Period	Relevance
Enterprise Application Modernization & Cloud Migration	Department of Defense (DoD) Component	2022-Present	Modernized 12 legacy Java/.NET applications to containerized microservices on AWS GovCloud (IL-5) ; implemented DevSecOps pipeline achieving cATO ; reduced deployment lead time from 6 months to 2 days ; \$4.2M annual cloud cost avoidance via FinOps
Federal Data Lakehouse & AI/ML Platform	Civilian Agency (Cabinet Level)	2021-2024	Built Petabyte-scale Lakehouse (S3 + Delta Lake, Redshift, Glue, Lake Formation); DBT models (500+); MLOps for fraud detection (SageMaker, MLflow, Feast); GenAI RAG for policy search (Bedrock, LangChain); FedRAMP High ATO
24/7 Cybersecurity Operations & Compliance Automation	Intelligence Community (IC) Element	2020-Present	SOC Type II operations (Splunk, Cortex XSOAR, MISP); OSCAL automation reducing A&A prep by 70% ; Continuous Control Monitoring for NIST 800-53 Rev. 5 (1,200+ controls); Zero Trust implementation (Zscaler, Entra ID, mTLS)
Citizen-Facing Digital Services & Accessibility Transformation	Federal Health Agency	2023-Present	Redesigned public-facing portal (Next.js, USWDS 3.0); achieved WCAG 2.1 AA conformance (VPAT/ACR); automated accessibility in CI (axe-core, Lighthouse); user research with assistive technology users ; 21st Century IDEA compliance
Agile Program Management & DevSecOps Transformation	DoD Joint Program Office	2022-2024	SAFe 6.0 implementation across 4 ARTs (200+ personnel); EVM integration (Jira Align); DORA metrics improvement: Deployment Frequency monthly -> daily , Lead Time weeks -> hours ,

			MTTR days -> minutes, Change Failure Rate 25% -> 3%
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Appendix B — Team & Leadership

OrbitAvanya Tech LLP's leadership team comprises professionals with extensive backgrounds in enterprise software delivery, cloud architecture, AI/ML systems, and government program management. Full team profiles and resumes are available upon request.